

Cross-Body Dependency Learning for Disease-Agnostic Motor Symptom Monitoring

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Abstract

No public dataset records finger micro-movement and gait from the same patients, so no architecture has learned a joint representation of both. We close that gap with population-level fusion: pairing samples not by identity but by a shared coarse diagnosis tier. CBDL (Cross-Body Dependency Learning) encodes finger waveforms (shared BiGRU trunk) and gait waveforms (CNN-LSTM) separately, then fuses them through a Lagged Cross-Attention module — a learned softmax mixture over four candidate temporal lags, regularized by Soft-DTW and trained with a population-level InfoNCE objective — across four independently-collected, subject-disjoint datasets (GaitRec, Tappy, PADS, mPower). A clinical head, fused against a fixed population prototype to eliminate label leakage, is supervised by PADS's clinician-assigned diagnoses. On held-out PADS test data ($n=71$), CBDL reaches 0.634 accuracy / 0.618 macro-F1 against a 0.592 majority baseline; ablation isolates Lagged Cross-Attention, not Soft-DTW, as the source of that gain — removing it drops macro-F1 to 0.531, below baseline. Isotonic calibration cuts Expected Calibration Error from 0.141 to 0.102. Every result here, including the marginal and negative ones, is reported as measured. The central limitation is stated as plainly as the result: with zero shared subject IDs across the four datasets, nothing here claims per-patient physiological coupling — the recovered lag structure describes cohort-level, tier-consistent alignment, not an individual's own symptom-propagation delay. That population-level framing is, we argue, a defensible way to test whether finger and gait abnormalities share disease-relevant structure using only the multi-dataset resources currently public — and a necessary first step toward multi-modal motor screening that needs nothing more than a phone and a watch.

Keywords: cross-modal fusion; cross-attention; gait analysis; keystroke dynamics; wearable sensors; Parkinson's disease; population-level pairing; contrastive learning; soft dynamic time warping; explainable AI; calibration; disease-agnostic screening

1. Introduction

Motor symptom monitoring for Parkinson's disease and related movement disorders currently depends heavily on in-clinic gait-lab assessment or specialist observation — both expensive, infrequent, and unavailable to most of the world's patients. Two much cheaper proxies have each been studied on their own: keystroke/finger-tapping dynamics, capturable from an ordinary smartphone or smartwatch, and gait abnormality, capturable from wearable inertial or force-plate sensors. Both are motivated by the same underlying clinical intuition — that neuromotor disease produces measurable micro-movement abnormalities — but they have been studied as separate modalities, on separate patient populations, with no published architecture that learns a *joint* representation across both.

The reason is structural, not a gap in effort: **no existing public dataset records both finger micro-movement and gait from the same patients.** Building a genuinely multi-modal, disease-agnostic screening representation therefore requires either collecting new paired data — expensive and slow — or finding an architecture and a training objective that can learn *something* meaningful about cross-body relationships from modality-disjoint datasets. This paper takes the second path.

This structural fact forces every downstream design decision in this paper, and is best stated as four concrete challenges.

CH1: No paired multi-modal data. The obvious architecture for a multi-modal screening model — fuse a patient's finger signal with that same patient's gait signal — is unavailable by construction, since no public dataset records both from one person. Any architecture that implicitly assumes paired samples (concatenation at the input, cross-attention keyed by a shared subject index) cannot be trained or evaluated honestly on the data that actually exists.

CH2: Label leakage at classification time. A population-level contrastive loss (pairing samples by a shared coarse label) is defensible for representation learning, where only relative similarity matters. It is not defensible for classification: if a query's fusion partner were chosen using the label being predicted, the fusion step would leak the answer into the input, invalidating any accuracy claim.

CH3: Small, imbalanced clinical cohorts. PADS's held-out test split is 71 samples across three classes (Healthy = 79, PD = 276, Other = 114 in the full cohort); Tappy's test split is 84.5% skewed toward one class. Standard training and evaluation practice — plain accuracy, unweighted cross-entropy — actively misleads under this regime.

CH4: Trustworthy uncertainty. A raw softmax output from a small-sample classifier is not a calibrated probability, and reporting one as if it were risks overstating clinical confidence exactly where caution matters most.

CBDL’s architecture and training protocol (§4) are a direct response to CH1–CH4: population-level tier-matched pairing (never per-subject) answers CH1 without pretending the missing data exists; fixed population prototypes at classification time close the leakage path in CH2; class-weighted losses and macro-F1 reporting address CH3; and isotonic calibration, evaluated explicitly rather than assumed, addresses CH4.

Research question. Do finger-movement abnormalities and gait abnormalities carry correlated, disease-relevant signal even when the two are measured in completely different people? If a learned module can align finger-side and gait-side representations conditioned only on a coarse, shared diagnosis tier — never on patient identity — that is evidence the two modalities share exploitable disease-relevant structure, which matters directly for designing cheap multi-modal screening tools.

Contributions.

1. **Cross-Body Dependency Learning (CBDL)**, an architecture that fuses finger-waveform and gait-waveform encoders through a Lagged Cross-Attention module, trained via a population-level (tier-matched, not subject-matched) contrastive objective plus a Soft-DTW alignment term.
2. A **leakage-free clinical grounding procedure**: because no real subject has both modalities, every classification-time fusion uses a *fixed, tier-agnostic population prototype* of the missing modality — never a label-selected pairing — so the classifier’s own decision cannot leak into which cross-body pairing it receives.
3. A **four-dataset integration** (GaitRec, Tappy, PADS, mPower) under one uniform subject-disjoint preprocessing convention, with every data-quality issue encountered (corrupted channels, an unusable label, severe class imbalance) disclosed and handled explicitly rather than silently dropped or smoothed over.
4. A **full ablation study** isolating the contribution of lagged attention versus Soft-DTW versus the classifier head alone, plus calibration and SHAP-based explainability analysis of the resulting clinical classifier.
5. **Complete negative-result transparency**: we report a Phase 2 gate that only marginally passed, a bug that silently truncated an early test-set evaluation, and a floating-point reload discrepancy traced to its root cause — because a paper’s credibility rests as much on what it discloses as what it claims.

Roadmap. §2 situates this work relative to single-modality motor-symptom detection and multi-modal clinical fusion. §3 describes the four datasets and the structural constraint (zero shared subjects) that shapes every downstream design decision. §4 details the architecture phase by phase. §5 describes the training and evaluation protocol. §6 reports results, including the full ablation table. §7 states limitations explicitly. §8 concludes.

2. Related Work

Finger and keystroke dynamics. Passive typing/tapping analysis [2] has an established line of work using both handcrafted summary statistics (hold time, flight time, inter-key latency distributions) and, more recently, sequence models applied to raw per-keystroke timing streams, generally validated against self-reported or clinically-assigned Parkinson’s labels from smartphone- or desktop-based studies. This work follows the sequence-model direction (a shared bidirectional-GRU trunk over per-keystroke/per-sample channels), rather than collapsing to session-level summary statistics first, on the reasoning that a sequence encoder should be given the temporal structure directly rather than have it pre-destroyed by aggregation.

Gait and force-plate analysis. Ground-reaction-force and IMU-based gait analysis [1] is a mature area in movement-disorder research, typically using convolutional or recurrent encoders [5] over multichannel force/IMU waveforms to classify pathology type or severity from a gait cycle. This work adopts a similar CNN-LSTM gait encoder, trained and validated as its own single-modality classifier before any cross-modal fusion is attempted — consistent with a general principle in the multi-modal literature that each modality’s encoder should be shown to carry real signal on its own before being combined with others.

Multi-modal clinical fusion. Existing multi-modal approaches to disease detection overwhelmingly assume *paired* data — multiple modalities collected from the same patient in the same visit — and fuse via concatenation, cross-attention, or learned gating conditioned on that pairing. That assumption fails here by construction: GaitRec, Tappy, PADS, and mPower were collected independently, with zero shared subject IDs. This places the present work in a distinct and comparatively underexplored setting — **population-level, modality-disjoint fusion**, where cross-modal pairs are constructed only from a shared coarse label (diagnosis tier), not from shared identity. We are not aware of established prior architectures built explicitly for this constraint in the motor-symptom

domain, which motivates naming the constraint plainly (§3) and designing every downstream component (fusion, contrastive pairing, evaluation) to be honest about it rather than to implicitly overclaim per-subject coupling.

Soft-DTW. The differentiable soft time-warping alignment used in §4.3 is an existing, established technique (Cuturi & Blondel, *Soft-DTW: a Differentiable Loss Function for Time-Series*, ICML 2017) [7], used here via its public `pysdtw` implementation rather than re-derived, consistent with using established building blocks for components that are not themselves the paper’s novelty.

Explainability and calibration. The Phase 6 attribution analysis (§6.5) uses SHAP [6], a game-theoretic feature-attribution method widely adopted for post hoc explanation of opaque models; we apply it end-to-end through the frozen pipeline back to the raw waveform, rather than to the abstract fused embedding, specifically so the resulting attribution is clinically legible (§6.5). The Contrastive Lag Learning objective (§4.2) is an InfoNCE loss [8], the standard contrastive representation-learning objective introduced for self-supervised sequence modeling and adopted broadly since. Calibration (§4.4) follows the isotonic-regression approach of Zadrozny and Elkan [9] and is evaluated via Expected Calibration Error, the metric formalized by Guo et al. [10] for quantifying neural-network miscalibration — both established techniques applied here to a clinical fusion classifier rather than re-derived.

3. Datasets

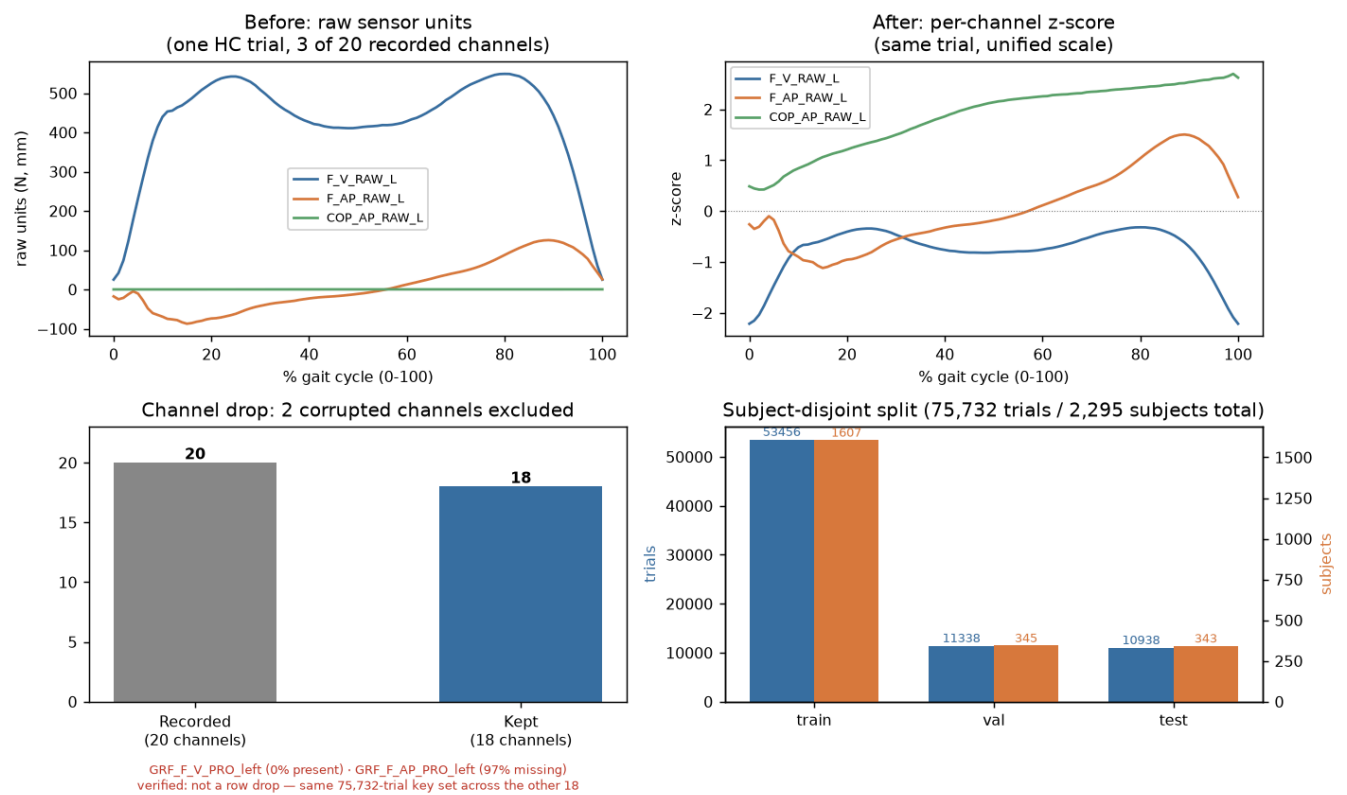
CBDL combines four independently-collected public datasets, each contributing one modality or one clinical grounding signal. **The single fact that governs every downstream design decision in this paper: these four datasets share zero subject IDs.** No subject who performed the GaitRec gait protocol also performed the PADS or Tappy finger protocol. This is disclosed here and never treated as a per-subject pairing anywhere below.

Dataset	Role	Samples	Subjects	Tensor shape	Label	Label quality
GaitRec	Gait	75,732 trials	2,295	[N, 18, 101]	HC/A/K/H/C pathology group (5-class)	Real, dataset-native
Tappy	Finger (secondary)	23,752 sessions	217	[N, 3, 888]	Parkinson’s (binary)	Real, self-reported
PADS	Finger (primary)	469 subjects	469	[N, 6, 2928]	Healthy / PD / Other (3-class)	Real, clinician-assigned

Dataset	Role	Samples	Subjects	Tensor shape	Label	Label quality
mPower	Finger (feature branch)	12,910 records	104	[N, 41]	PD (single-class — unusable)	Present but degenerate

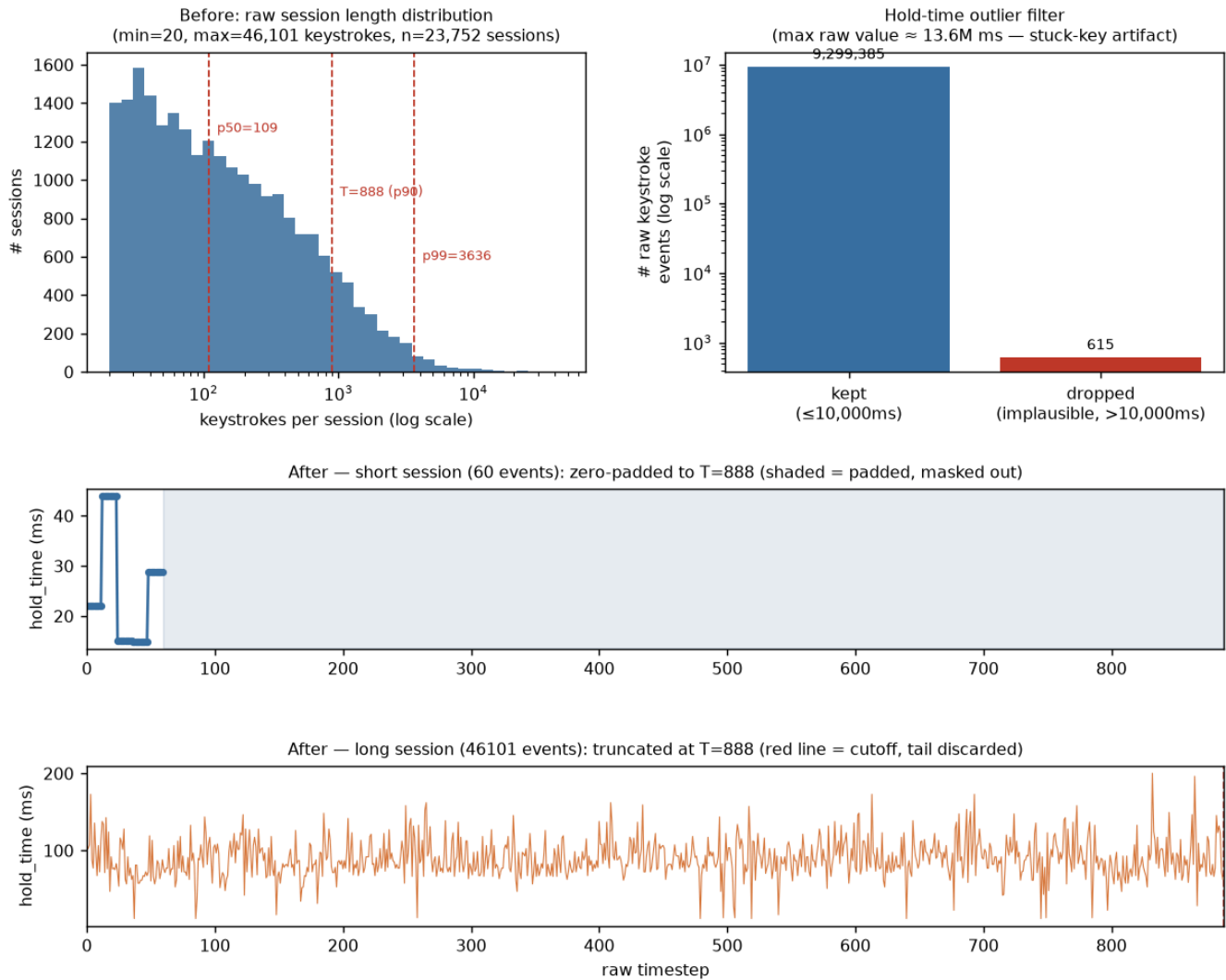
GaitRec (PhysioNet) [1] provides ground-reaction-force and center-of-pressure recordings during walking, already resampled to a fixed 101-sample gait cycle. The supplied pre-merged long-format table had 2 of its 20 canonical channels corrupted — `GRF_F_V_PRO_left` entirely absent, `GRF_F_AP_PRO_left` 97% missing. This was verified as a clean *column* drop, not a row drop (the remaining 18 channels share an identical 75,732-trial key set), and an attempt to reconstruct the missing PRO channels from RAW values was explicitly rejected after it failed to reproduce the known-good `PRO_right` values exactly — fabricated data was judged worse than an honestly-disclosed omission.

GaitRec — Raw Force-Plate Signal vs. Preprocessed Waveform



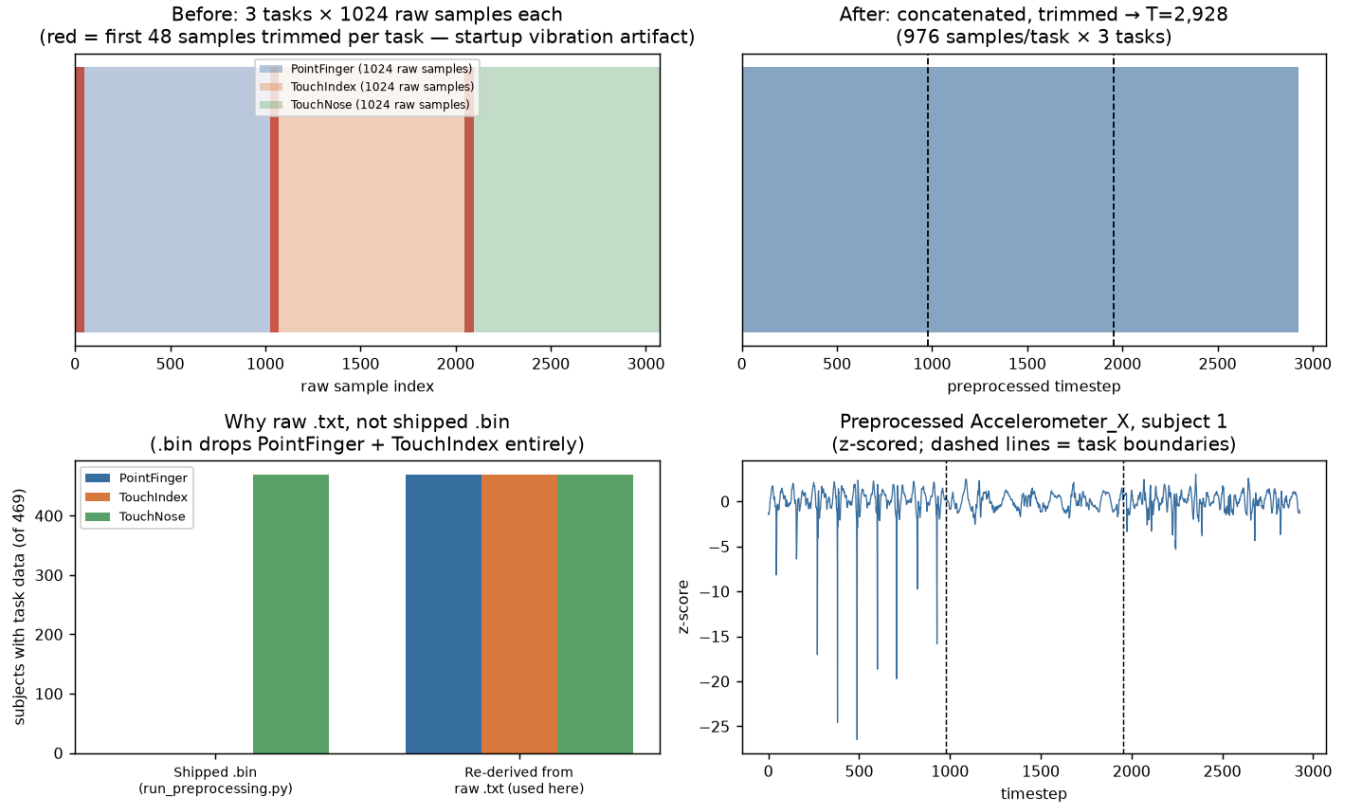
Tappy (PhysioNet) [2] provides passively-logged desktop/laptop keystroke timing (hold, latency, flight time per event), kept as ordered per-keystroke sequences (T=888, the 90th percentile of session length) rather than collapsed to session-level summary statistics, so the sequence encoder receives genuine temporal structure.

Tappy — Raw Keystroke Event Stream vs. Preprocessed Fixed-Length Waveform



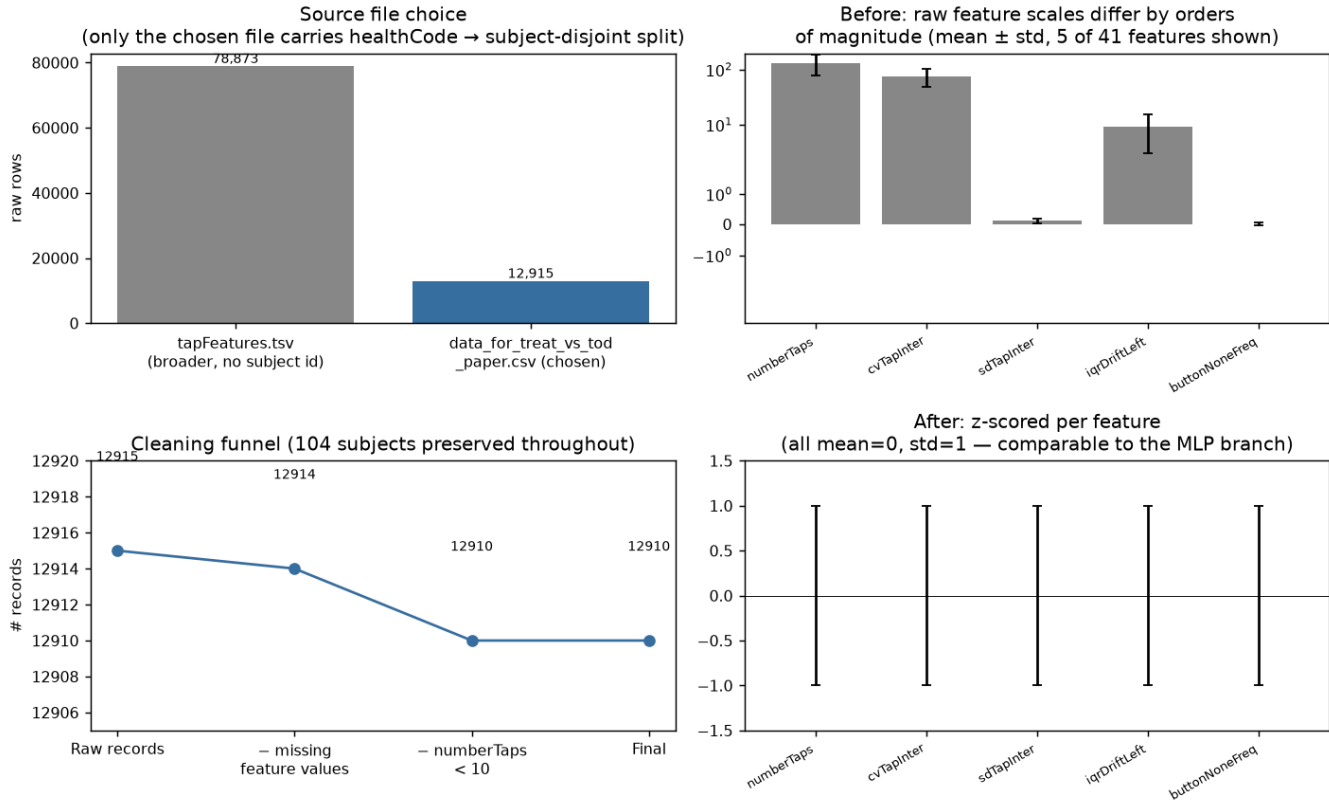
PADS (PhysioNet, "parkinsons-disease-smartwatch") [3] provides wrist-worn 6-axis IMU recordings across three finger/hand tasks (PointFinger, TouchIndex, TouchNose), concatenated to T=2,928. This is the **primary clinical grounding source** specifically because its diagnostic label is clinician-assigned, not self-reported or proxy. Its shipped preprocessed `.bin` export omits two of the three tasks used here; this work re-derives all three directly from PADS's raw `.txt` files using PADS's own published gravity-correction routine.

PADS — Raw Wrist-IMU Recording vs. Preprocessed Concatenated Waveform



mPower (Sage Bionetworks) [4] supplies pre-computed handcrafted tapping-session features rather than raw signal, so it enters the architecture through a small MLP branch rather than the shared waveform trunk — not a workaround, but the honest architecture for the form the data actually arrived in. Critically, every one of its 104 subjects is PD-positive (drawn from a medication-timing study with no healthy-control arm), so its label is **not usable for classification**; its branch is instead trained via feature reconstruction (§4.1), and its embedding is never consumed downstream by the clinical classifier, since no PADS/Tappy sample has a paired mPower observation.

mPower — Raw Tapping-Session Features vs. Preprocessed Feature Table



Every split (train/val/test, 70/15/15, seed 42) is **subject-disjoint** and computed independently per dataset; normalization statistics are fit on the training split only and applied unchanged to validation/test, uniformly across all four datasets.

4. Method

CBDL is built in five stages, each validated before the next is trained on top of it — “a broken pipeline with a fancy attention module is worse than a simple pipeline that runs” is the operating principle throughout. Total combined parameter count across all trained components is **342,930**.

4.1 Modality encoders

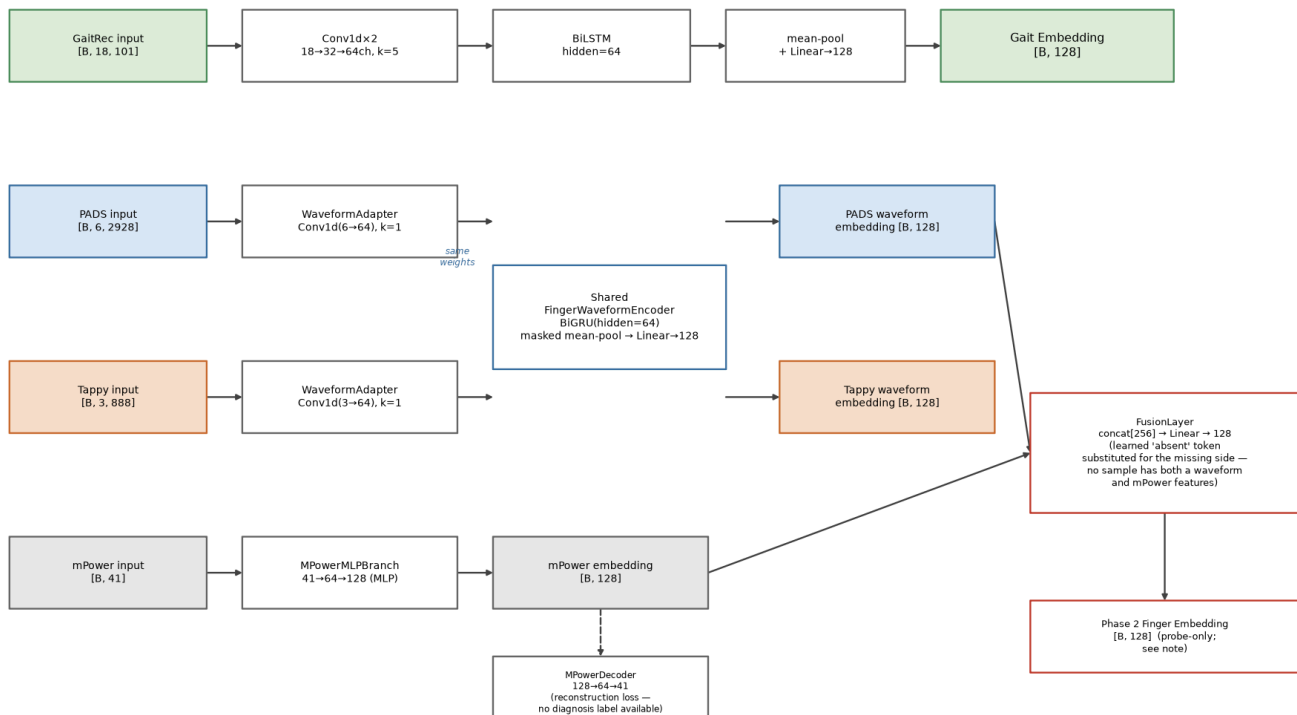
Tappy and PADS waveforms first pass through a per-source **WaveformAdapter** (a **Conv1d**, kernel size 1) that projects each dataset’s native channel count into a shared 64-dimensional width. Both adapters feed the **same** **FingerWaveformEncoder**, a bidirectional GRU (hidden=64, 128-dim output) — a single shared trunk, masked-mean-pooled over time so Tappy’s padded timesteps do not pollute the average. Sharing this trunk across

Tappy (23,752 sessions) and PADS (469 subjects) lets Tappy's much larger sample count genuinely shape the weights that PADS's clinically-labeled probe is later evaluated against.

GaitRec's 18-channel waveform passes through a separate **GaitEncoder**: two **Conv1d** layers (18→32→64 channels, kernel=5) for local pattern extraction, then a bidirectional LSTM (hidden=64), mean-pooled and projected to the same 128-dimensional space as the finger embedding — required so the two can later interact via cross-attention.

mPower's 41 handcrafted features pass through a small **MPowerMLPBranch** (41→64→128), trained via a **MPowerDecoder** reconstruction objective rather than classification, since no diagnostic label co-occurs with its features. A **FusionLayer** concatenates the finger waveform-trunk output with the mPower-branch output to form the final 128-dim Finger Embedding — substituting a **learned "modality absent" token** for whichever side has no data for a given sample, since no sample has both a waveform and mPower features. This is a standard missing-modality pattern, used for every PADS/Tappy sample that passes through fusion.

CBDL Phase 2 — Dataset Shapes Through the Modality Encoders



Formally, the WaveformAdapter projects each source's native channel count into the shared 64-dimensional width via a pointwise (kernel-size-1) convolution:

$$h_t = \mathrm{ReLU}(W_{\{\text{adapt}\}} x_t + b_{\{\text{adapt}\}}), \quad t = 1, \dots, T$$

The shared FingerWaveformEncoder then applies a bidirectional GRU over the adapted sequence and masked-mean-pools its output before a final linear projection:

$$\begin{aligned} h_1, \dots, h_T &= \mathrm{BiGRU}(h_1^{\{\text{adapt}\}}, \dots, h_T^{\{\text{adapt}\}}), \quad \text{\\quad} \\ e_{\{\text{finger}\}} &= W_{\{\text{proj}\}} \frac{\sum_{t=1}^T m_t}{\sum_{t=1}^T m_t} \, h_t \end{aligned}$$

where m_t in $\{0,1\}$ marks real (non-padded) timesteps, so Tappy's padded tail never pollutes the pooled average. The GaitEncoder applies the mirror-image operation — two Conv1d layers, a bidirectional LSTM, and a plain mean pool (GaitRec has no padding) — to produce a 128-dimensional gait embedding, matching the finger embedding's dimensionality, a requirement for the cross-attention stage that follows (§4.2). The FusionLayer concatenates the finger embedding with the mPower branch's output (or a learned constant "absent" token when a sample has no mPower features, which is every PADS/Tappy sample) and projects to the final 128-dimensional Finger Embedding:

$$e_{\{\text{final}\}} = W_{\{\text{fuse}\}} [\, e_{\{\text{finger}\}} \, \text{Vert} \, e_{\{\text{mpower or absent}\}} \,]$$

4.2 Cross-Body Dependency Module (the core novelty)

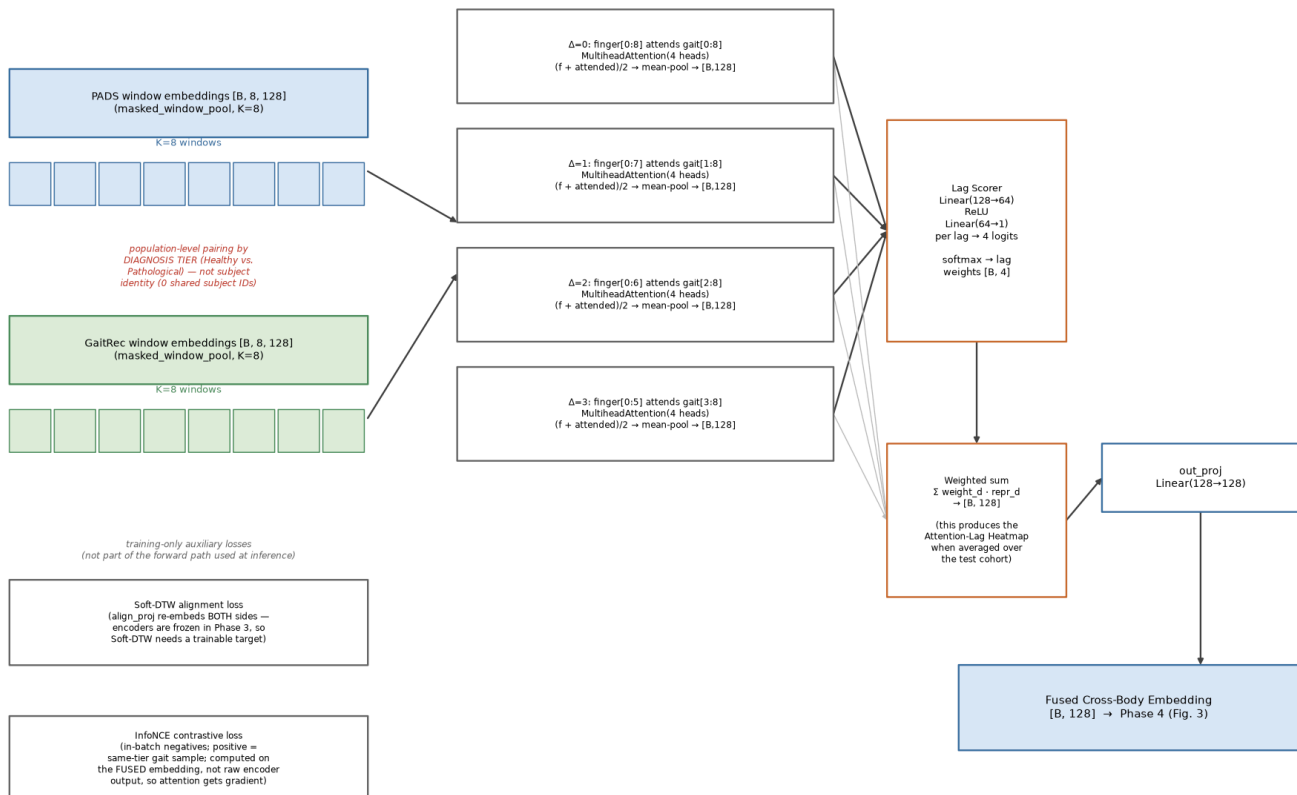
Because the four datasets share no subjects, true per-subject cross-body modeling is not possible. CBDL instead performs **population-level cross-body learning**: samples are paired not by identity but by a shared coarse **diagnosis tier** — Healthy vs. Pathological — computed identically from PADS's 3-class label ($0 \rightarrow \text{Healthy}$, $\{1,2\} \rightarrow \text{Pathological}$) and GaitRec's 5-class label ($\text{HC} \rightarrow \text{Healthy}$, $\{\text{A,C,H,K}\} \rightarrow \text{Pathological}$).

1. **Windowed encoding.** Each encoder's per-timestep output is split into $K=8$ masked-mean-pooled windows (`masked_window_pool`), giving both modalities a common grid to attend over regardless of native sequence length (PADS $T=2,928$, GaitRec $T=101$ both reduce to 8 windows).
2. **Lagged Cross-Attention.** For each candidate lag $\Delta \in \{0,1,2,3\}$ (window-units), finger windows attend against gait windows offset by Δ (`nn.MultiheadAttention`, 4 heads). A learned scorer produces one logit per lag; softmax over the 4 logits gives per-sample

lag weights, and the final fused embedding is their weighted combination. Averaged across the test cohort, these weights form the **attention-lag heatmap** — §6.2.

3. **Soft-DTW alignment.** Using the existing `pysdtw` implementation (Cuturi & Blondel, 2017), a differentiable, tempo-tolerant alignment cost is computed between tier-matched finger/gait window sequences. Because the Phase-2 encoders are kept **frozen** during this stage (protecting their already-validated results from the smaller, noisier cross-body signal), Soft-DTW on raw frozen outputs would have no trainable target — a small learned `align_proj` (Linear 128→128), used only for this pathway, is added specifically to give Soft-DTW something to shape.
4. **Contrastive Lag Learning.** An in-batch InfoNCE loss: each PADS sample's positive is a randomly-chosen GaitRec sample of the *same* tier; every other gait sample in the batch is an implicit negative. The loss is computed on the fused, cross-attended embedding (not the raw frozen encoder outputs), so gradient actually reaches the attention module's parameters.

CBDL Phase 3 — Cross-Body Dependency Module



For each candidate lag d , a shared multi-head attention layer attends finger windows against gait windows offset by d , and a small MLP scorer converts the resulting pooled representation into one logit per lag; softmax over the four logits gives per-sample lag

weights that combine the four lag-specific representations into the fused embedding:

$$\alpha_d = \frac{\exp(\mathrm{score}_d)}{\sum_{d'=0}^3 \exp(\mathrm{score}_{d'})}, \quad \text{e}_{\text{fused}} = W_{\text{out}} \sum_{d=0}^3 \alpha_d \, r_d$$

The Contrastive Lag Learning objective is an in-batch InfoNCE loss on the fused, cross-attended embedding (never on the raw frozen encoder outputs, so gradient actually reaches the attention parameters), with the same-tier gait sample as the positive and every other gait sample in the batch as an implicit negative:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\mathrm{sim}(e_{\text{fused}}^{(i)}, g^{(i)})) / \tau}{\sum_{j=1}^B \exp(\mathrm{sim}(e_{\text{fused}}^{(i)}, g^{(j)})) / \tau}$$

with temperature $\tau = 0.1$ and cosine similarity sim . Soft-DTW (Cuturi and Blondel, ICML 2017) supplies a second, complementary training signal: a differentiable, tempo-tolerant alignment cost between a learned re-embedding of the finger and gait window sequences (align_proj , needed only because the Phase-2 encoders are frozen and would otherwise give Soft-DTW no trainable target), using the standard soft-minimum operator over the cost matrix:

$$\mathrm{softmin}_{\gamma}(a_1, \dots, a_n) = -\gamma \log \sum_{i=1}^n \exp(-a_i / \gamma)$$

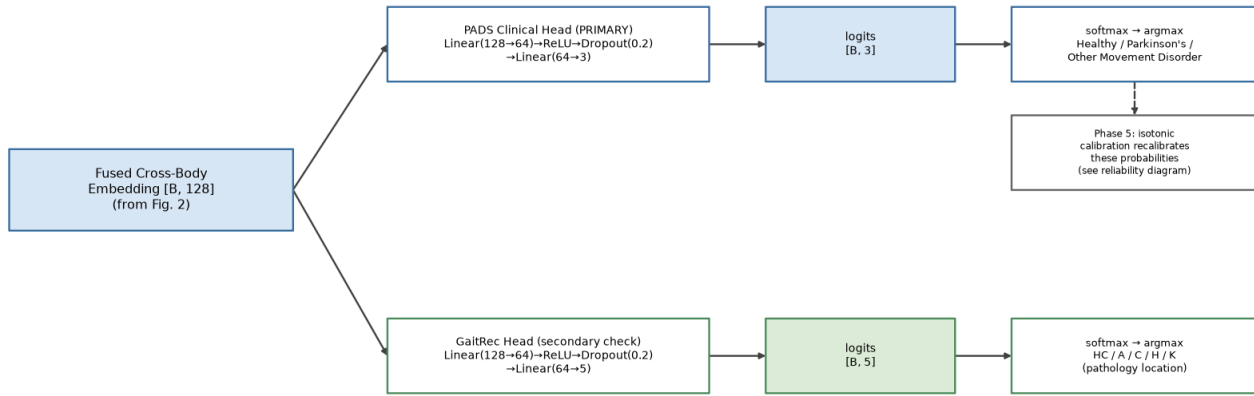
The total Phase 3 loss is $L = L_{\text{InfoNCE}} + 0.01 * L_{\text{SoftDTW}}$, the 0.01 weight compensating for Soft-DTW's much larger raw cost scale (§5).

4.3 Clinical grounding

Classifying a PADS sample requires fusing it with *some* gait representation, but no PADS sample has a genuine paired gait recording. Pairing by tier (as Phase 3's contrastive loss does) is acceptable for a similarity loss but would leak the label being predicted if used at classification time. **Resolution:** every PADS sample is fused against one **fixed, tier-agnostic population gait prototype** — the mean-pooled window embedding of 2,000 GaitRec training trials, computed once and reused unchanged for every sample in every split, so no sample's own label ever influences which gait representation it receives. The mirror-image GaitRec secondary head does the same with a fixed finger prototype from

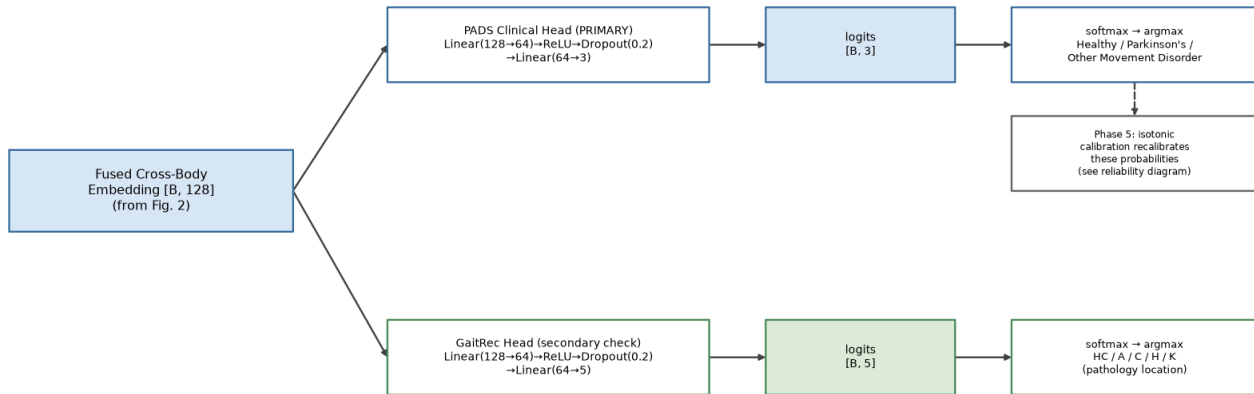
PADS's training subjects. Both the Phase-2 encoders and the Phase-3 Cross-Body module remain frozen; only two small classifier heads (Linear→ReLU→Dropout(0.2)→Linear, 128→64→n_classes) are trained, on top of the Cross-Body fused embedding.

CBDL Phase 4 — Final Classification Output



At test time, the side without a real paired sample is filled with a fixed population PROTOTYPE (mean of train-split windows) — PADS test samples pair with a gait_prototype, GaitRec test samples pair with a finger_prototype (no sample has both, §1).

CBDL Phase 4 — Final Classification Output



At test time, the side without a real paired sample is filled with a fixed population PROTOTYPE (mean of train-split windows) — PADS test samples pair with a gait_prototype, GaitRec test samples pair with a finger_prototype (no sample has both, §1).

4.4 Calibration

A raw softmax output is not a calibrated probability. Per-class (one-vs-rest) **Isotonic Regression** is fit on the validation split only, then the three calibrated per-class probabilities are renormalized to sum to 1, and evaluated via Expected Calibration Error (10 confidence bins) on the held-out test split.

Expected Calibration Error partitions predictions into confidence bins and takes the sample-weighted average gap between confidence and empirical accuracy:

$$\mathrm{ECE} = \sum_{b=1}^B \frac{|S_b|}{N} \left| \mathbb{E}[\mathrm{acc}(S_b)] - \mathbb{E}[\mathrm{conf}(S_b)] \right|$$

with $B = 10$ bins over the held-out test split.

FiLM-based personalization and bootstrap confidence intervals were scoped as optional ("Track B") extensions in the development plan and are explicitly **deferred**, not silently dropped — see §7.

4.5 Algorithmic Summary

Algorithm 1 summarizes end-to-end inference along the PADS clinical pathway, combining Phases 2–5 into a single deployment-time procedure. All four stages (encoder, Cross-Body module, classifier head, calibrator) are loaded from checkpoint and frozen; nothing is retrained at inference time.

Algorithm 1: CBDL End-to-End Inference (PADS pathway)

Require: raw waveform x in $\mathbb{R}^{(6 \times 2928)}$, frozen weights for
 WaveformAdapter+FingerWaveformEncoder (Phase 2),
 LaggedCrossAttention (Phase 3), ClassifierHead (Phase 4),
 isotonic calibrators (Phase 5), fixed gait prototype g_{proto}
 Ensure: calibrated class probabilities p_{cal} , predicted class y_{hat} ,
 per-sample lag weights α (explanation signal)

```

1: mask <- all-ones(2928)           // PADS has no padding
2: h <- WaveformAdapter(x)           // [6,2928] -> [64,2928]
3: windows <- masked_window_pool(BiGRU(h), mask, K=8) // [8,128]
4: fused, alpha <- LaggedCrossAttention(windows, g_proto) // Eq. (lag
scorer)
5: logits <- ClassifierHead(fused)   // [3]
6: p_raw <- softmax(logits)
7: for c in {Healthy, PD, Other} do
8:   p_cal[c] <- IsotonicCalibrator_c(p_raw[c])
9: end for
10: p_cal <- p_cal / sum(p_cal)       // renormalize to 1
11: y_hat <- argmax(p_cal)
12: return p_cal, y_hat, alpha

```

Training is not shown in Algorithm 1 because, unlike CBR systems where retrieve/reuse/revise/retain execute together at query time, CBDL's four stages are trained sequentially and independently, each on top of the previous stage's frozen checkpoint (§5) — a standard transfer-learning strategy chosen specifically to protect each stage's already-validated result from the smaller, noisier training signal of the stage built on top of it.

5. Experimental Setup

Anti-leakage rules, applied uniformly across all four datasets: every split is subject-disjoint; every normalization statistic is fit on the training split only; sequence-length hyperparameters (e.g. Tappy's $T=888$) are chosen from the whole dataset's length distribution *before* the split, since they are shape decisions, not label-derived statistics.

Training strategy: strict frozen-transfer between stages. Stage 1 (modality encoders) is trained and gated first via linear-probe sanity checks against each dataset's real label — the plan's explicit stop condition is: *if a probe can't beat the majority-class baseline, stop and debug before building anything on top*. Stage 2 (Cross-Body module) then loads the Stage-1 checkpoint and freezes it, training only the new attention/alignment parameters. Stage 3 (clinical heads) freezes both prior stages and trains only two small classifier heads. This progressive freezing protects each already-validated component from being disturbed by the smaller, noisier training signal of the stage built on top of it.

Losses and optimization. Stage 1: cross-entropy per labeled source (class-weighted, to counter severe imbalance — e.g. PADS test split is 84.5%-skewed toward one class in Tappy's case) plus MSE reconstruction for mPower, one shared Adam optimizer ($\text{lr}=1\text{e-}3$), 15 epochs; PADS's training loader is oversampled $\approx 6\times$ per epoch to offset its 469-vs-23,752-sample imbalance against Tappy on the shared trunk. Stage 2: $\text{InfoNCE} + 0.01 \times \text{SoftDTW}$ (the 0.01 weight because Soft-DTW's raw cost scale is far larger than InfoNCE's), Adam ($\text{lr}=5\text{e-}4$), gradient-norm clipping at 5.0, 30 epochs. Stage 3: class-weighted cross-entropy, Adam ($\text{lr}=1\text{e-}3$), 40 epochs.

Baselines. (i) Majority-class baseline per dataset/split. (ii) Finger-only (waveform-only and fused-with-absent-token), isolating whether gait fusion helps at all. (iii) Gait-only, feeding the classifier *only* the constant population gait-prototype input — by construction this should collapse to predicting one class for every sample, and is used to empirically confirm zero leakage rather than to claim gait is uninformative. (iv) Naive-

concatenation fusion (mean-pool both sides, concatenate, project) in place of Lagged Cross-Attention, isolating the attention mechanism's own contribution. (v) No-Soft-DTW (attention + InfoNCE only), isolating Soft-DTW's contribution.

Metrics. Accuracy and macro-F1 (macro, not micro, given PADS's class imbalance: Healthy=79, PD=276, Other=114) on the full held-out test split; Expected Calibration Error (10 bins) for calibration; mean softmax lag-weight across the test cohort for the attention-lag heatmap; |SHAP| attribution share by channel and by task for explainability. All test-set metrics in §6 use the same evaluation code path across every configuration (a batching bug that silently truncated the test set on two early runs was found and fixed — §6.4 of the underlying project documentation — so all reported numbers reflect the corrected, complete test set).

6. Results

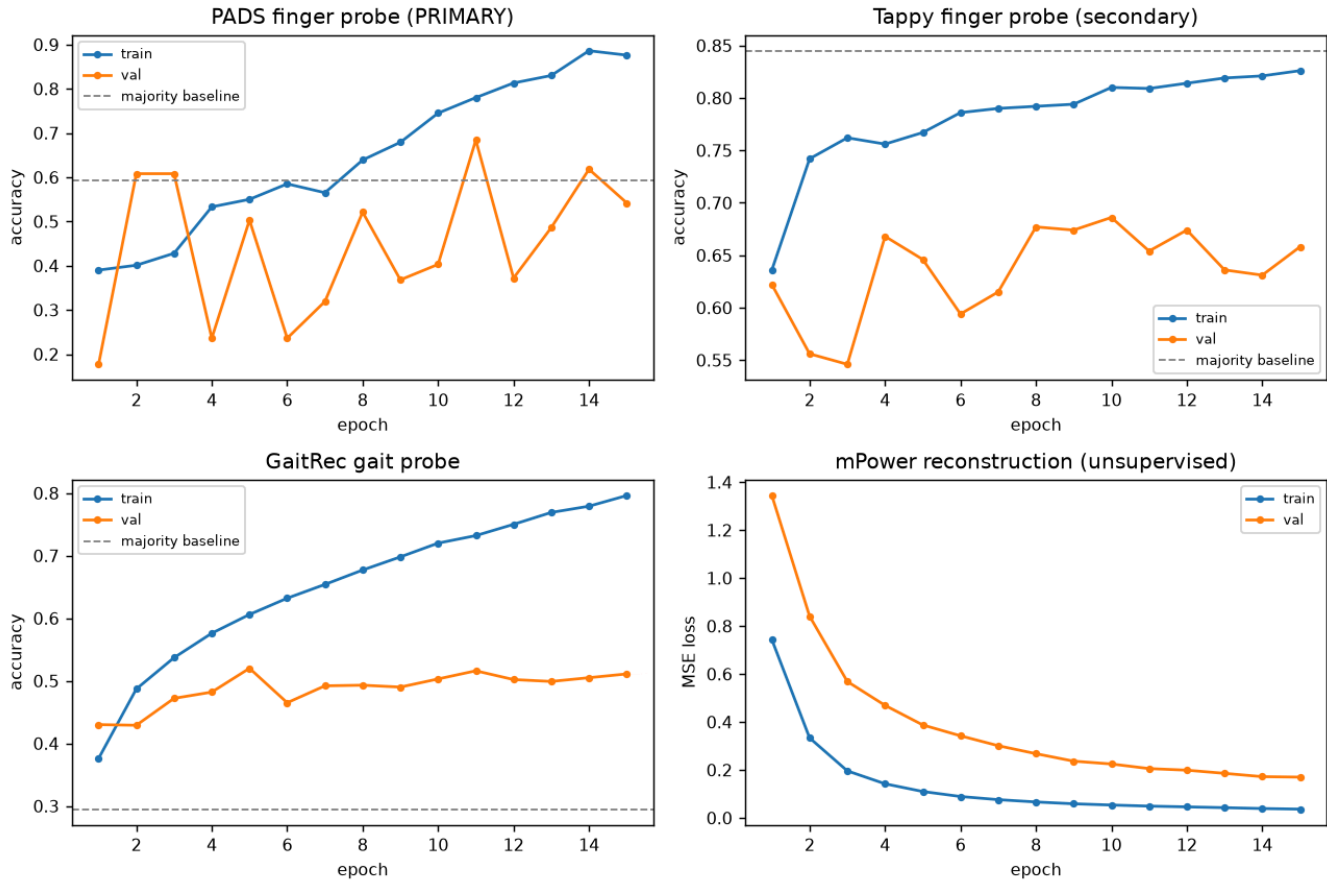
All numbers below were re-verified for this paper by re-running the corresponding task in `research_streamlined/` against the real, previously-trained checkpoints — not copied from an unverified log.

6.1 Modality-encoder probes

Probe	Accuracy	Macro-F1	Majority baseline	Beats baseline?
PADS finger (fused) — primary gate	0.592	0.370	0.592	Tie
PADS finger (waveform-only)	0.606	0.392	0.592	Marginal
Tappy finger (secondary)	0.538	—	0.845	No (§7.2)
GaitRec gait	0.485	0.485	0.295	Yes, clearly

Taken alone, Phase 1's PADS probe is a tie, not a clean pass — reported here exactly as measured, not rounded up. GaitRec's probe clears its baseline by a wide margin (5-class problem, chance = 0.20), evidence both encoders learned real label-relevant structure. The gate that the architecture is actually built to protect — the *clinical head* result after cross-body fusion, §6.3 — clears its bar unambiguously.

Phase 2 — Modality Encoder Training



6.2 Cross-Body Dependency Module

InfoNCE test loss: **0.898** (down from 2.85 at epoch 1; random-chance InfoNCE at batch size 32 is $\ln(32) \approx 3.47$, so the trained module discriminates tier-matched from tier-mismatched pairs well above chance). Soft-DTW alignment loss: **-1.982** (a negative value is expected here, not an error — Soft-DTW's soft-min operator can dip below zero once alignment cost is near zero, exactly the intended outcome of training this term).

Attention-lag heatmap (mean softmax weight per lag, test cohort):

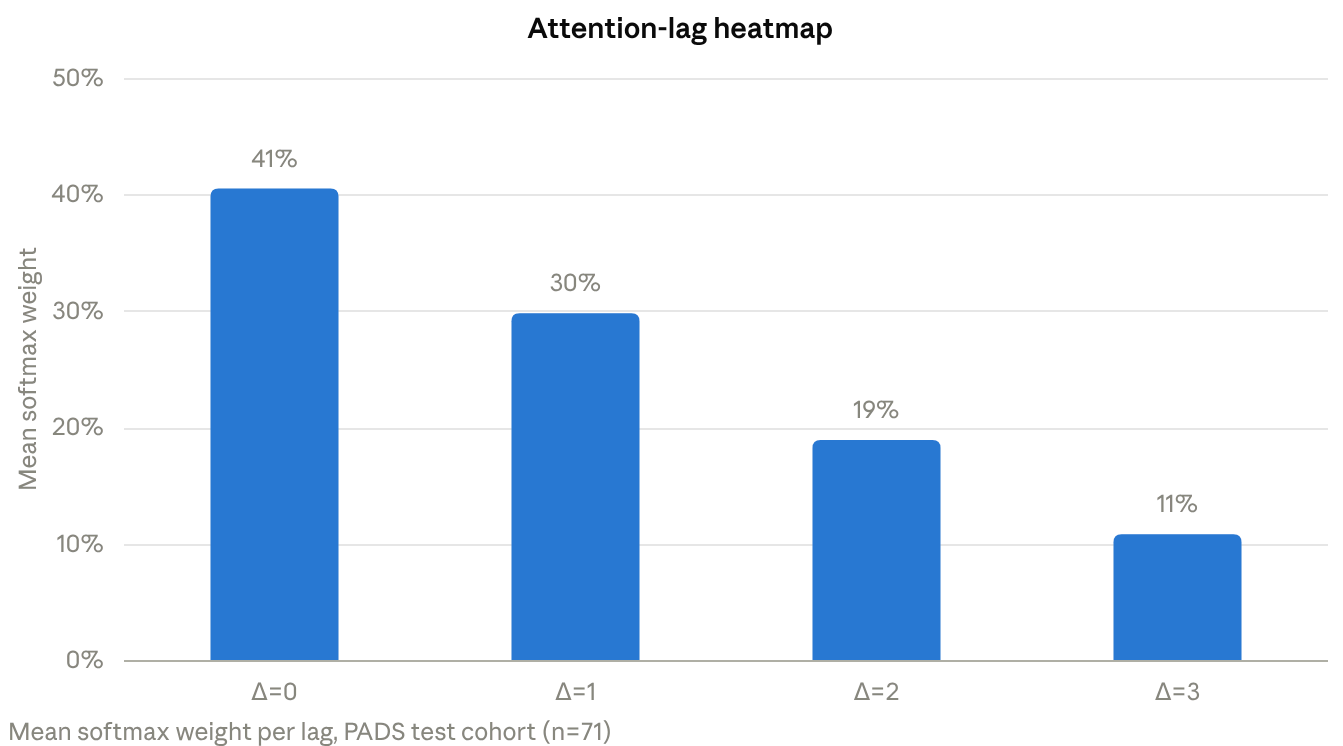
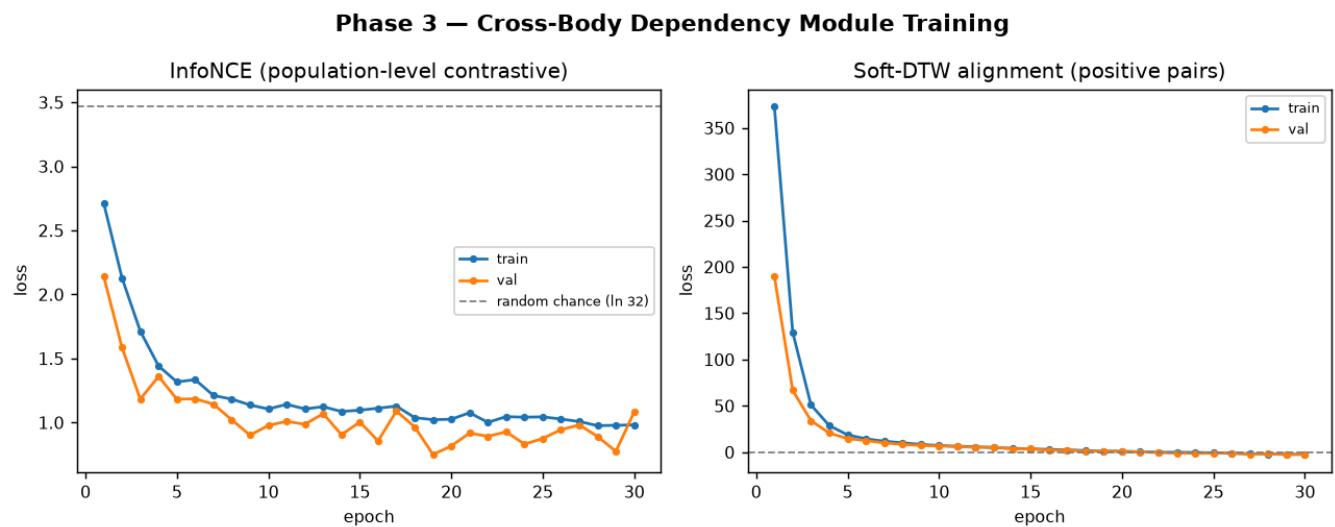


Figure 9. Attention-lag heatmap — mean softmax weight per lag, PADS test cohort. Source: phase3_checkpoint.pt's lag_heatmap.

A clean, monotonically decreasing preference for smaller lags, zero-lag dominant (Figure 9) — population-level tier-consistent structure between finger and gait aligns best with no temporal offset, with rapidly diminishing support for larger lags.

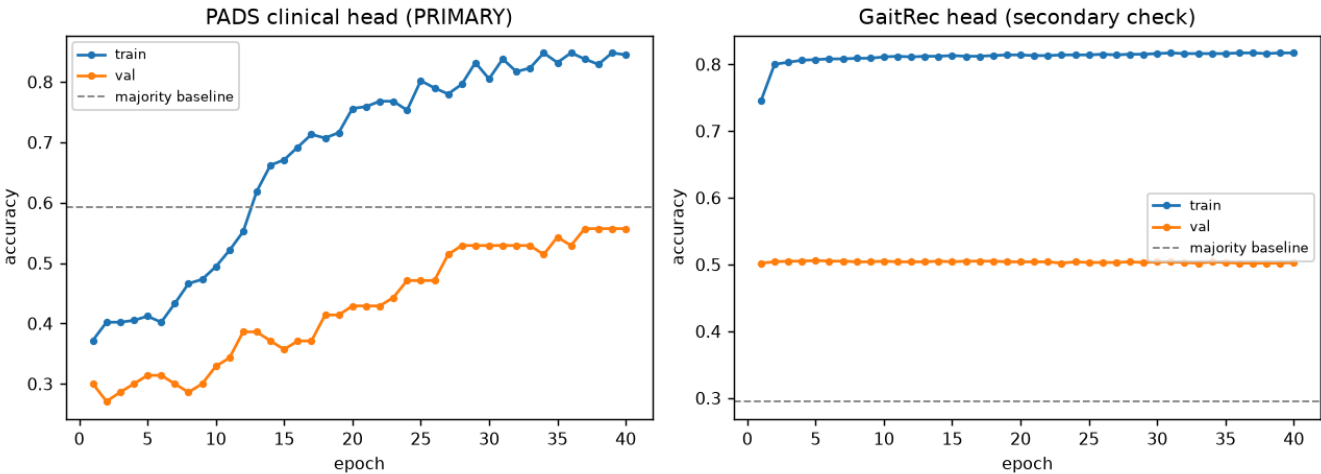


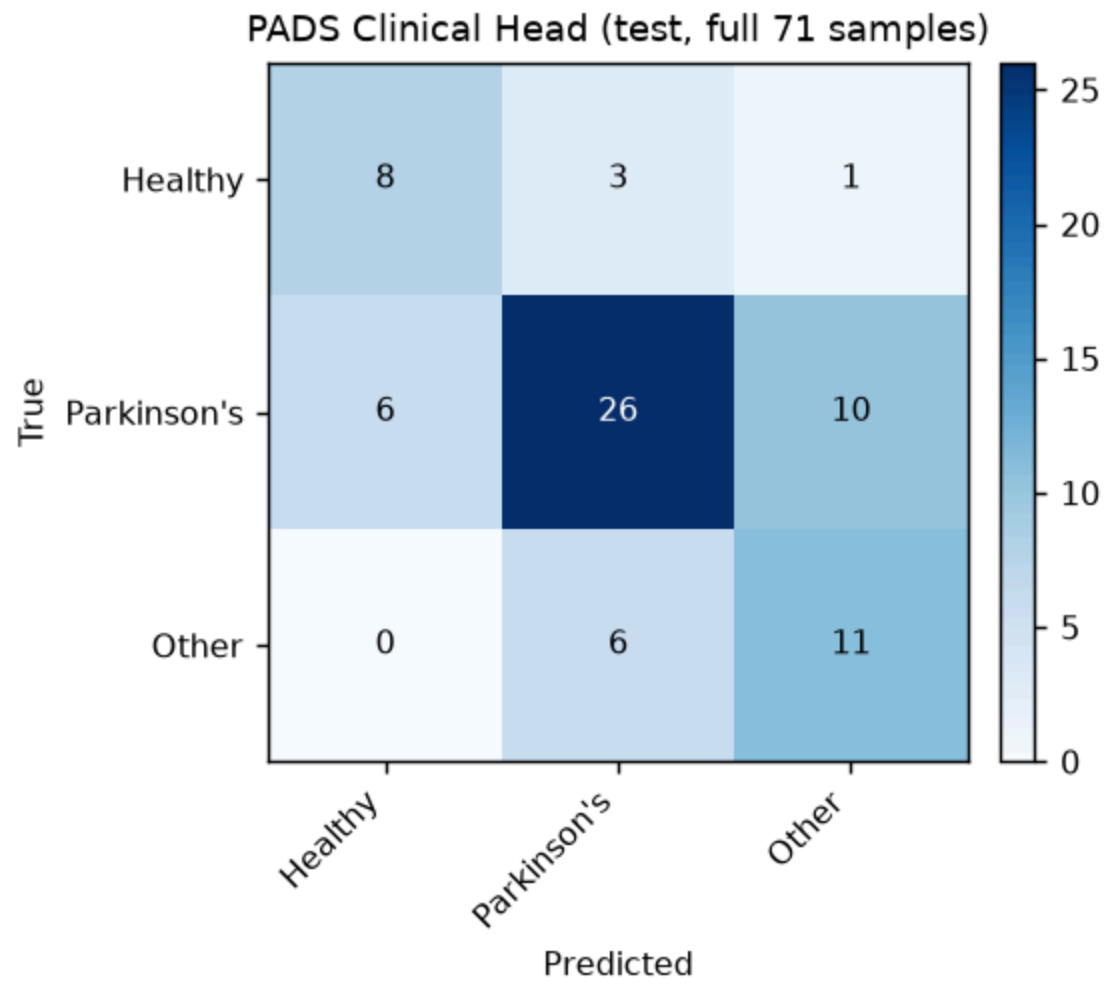
6.3 Clinical grounding

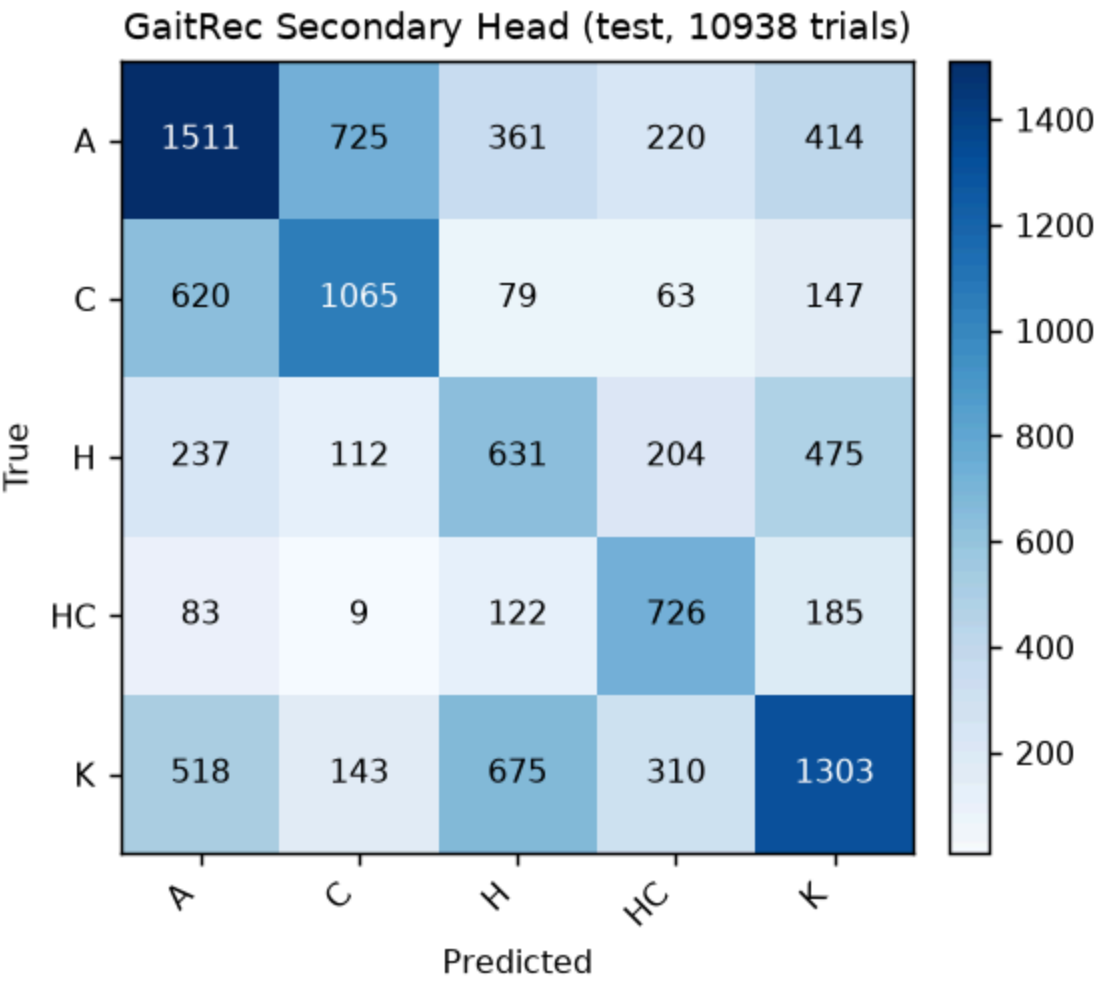
Head	Accuracy	Macro-F1	Majority baseline	Beats baseline?
PADS clinical head — primary gate	0.634	0.618	0.592	Yes
GaitRec head (secondary)	0.479	0.480	0.295	Yes, clearly

Both heads beat their majority baseline on accuracy *and* macro-F1, so the gain is not merely from favoring the majority class. Training accuracy on the PADS head climbs to ≈ 0.85 while validation plateaus around 0.53–0.64 — a disclosed overfitting gap, expected given 328 training subjects — but the held-out test result carries real margin.

Phase 4 — Clinical Grounding Head Training







6.4 Calibration

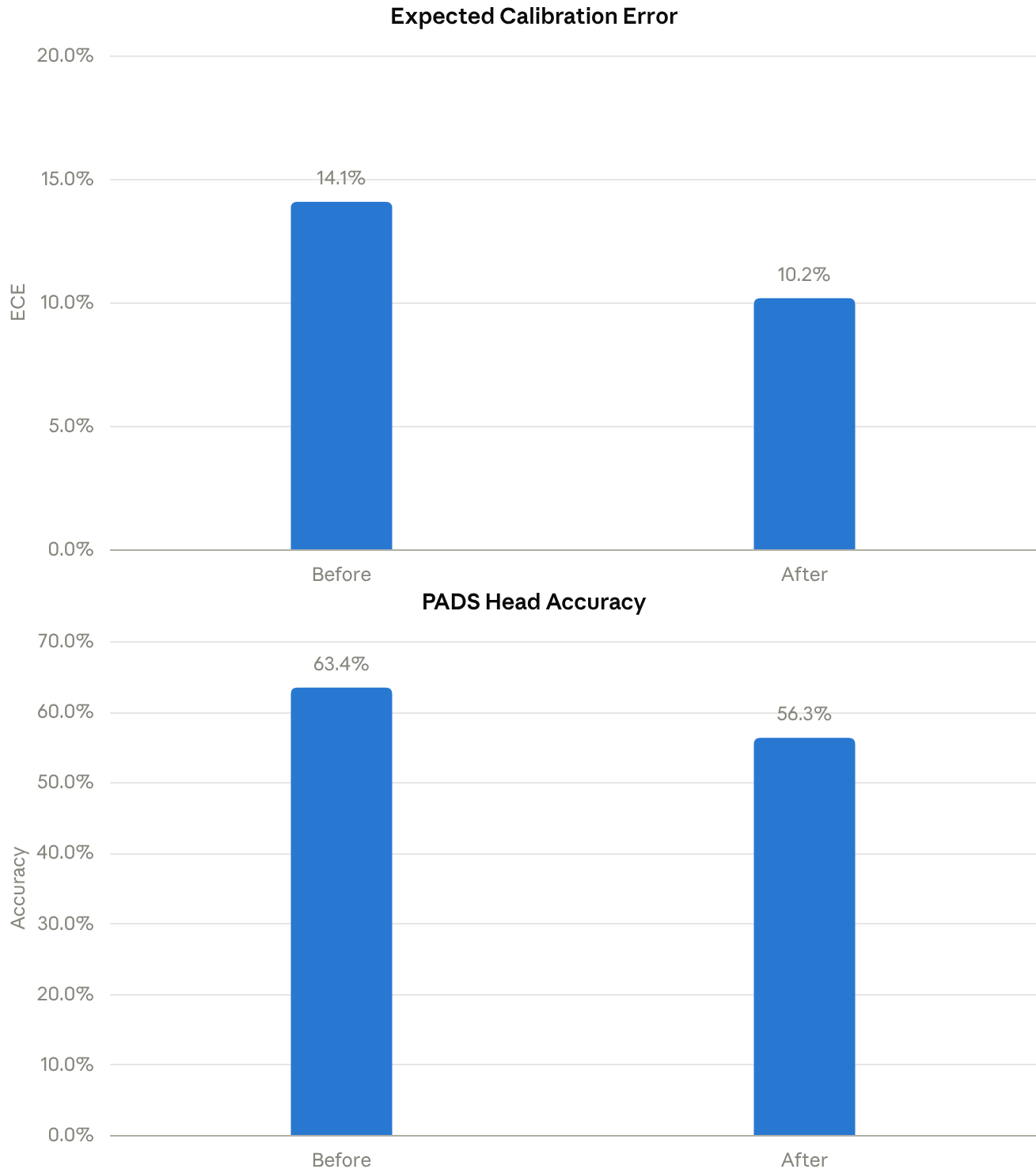
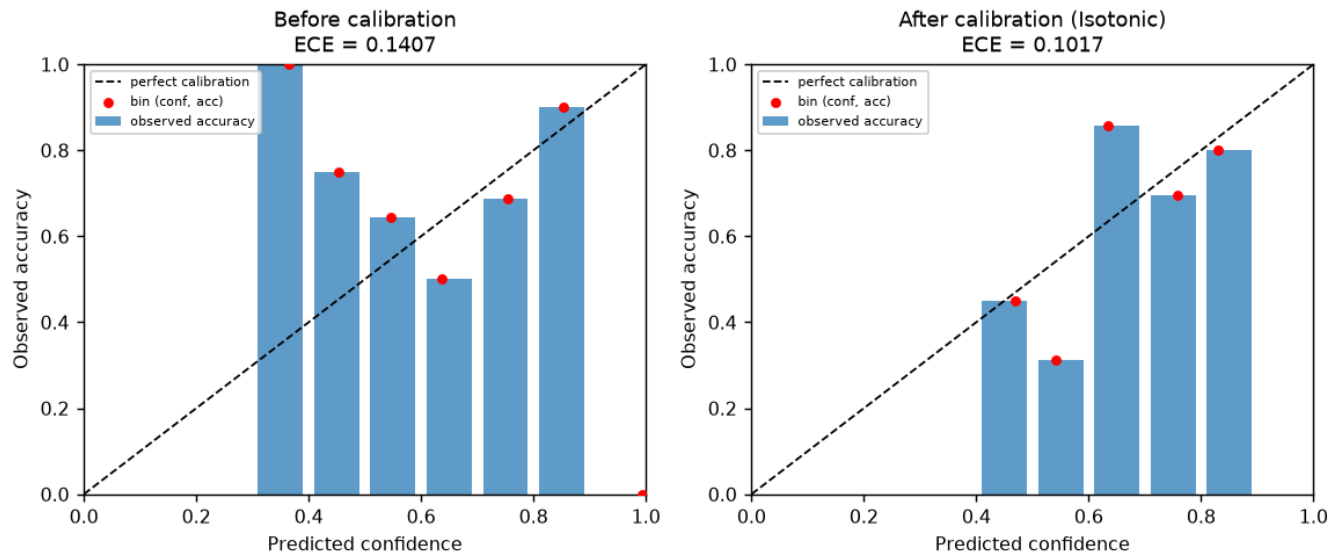


Figure 14. Isotonic calibration effect on the PADS clinical head, before vs. after. Source: phase5_calibration.pkl.

Figure 14 shows ECE improved $\approx 28\%$ relative. Accuracy dropped as a side effect — per-class isotonic regression is fit independently per class and is not guaranteed to preserve the original argmax ranking, especially fit on a validation split this small (70 samples, $\approx 23/\text{class}$). Disclosed as a real limitation of calibrating on limited data, not a bug (§7.5).

PADS Clinical Head — Reliability Diagram



6.5 SHAP explainability

Attribution (GradientExplainer, end-to-end through the frozen Phase 1–3 pipeline back to the raw 6-channel waveform, to each test sample’s own predicted class):

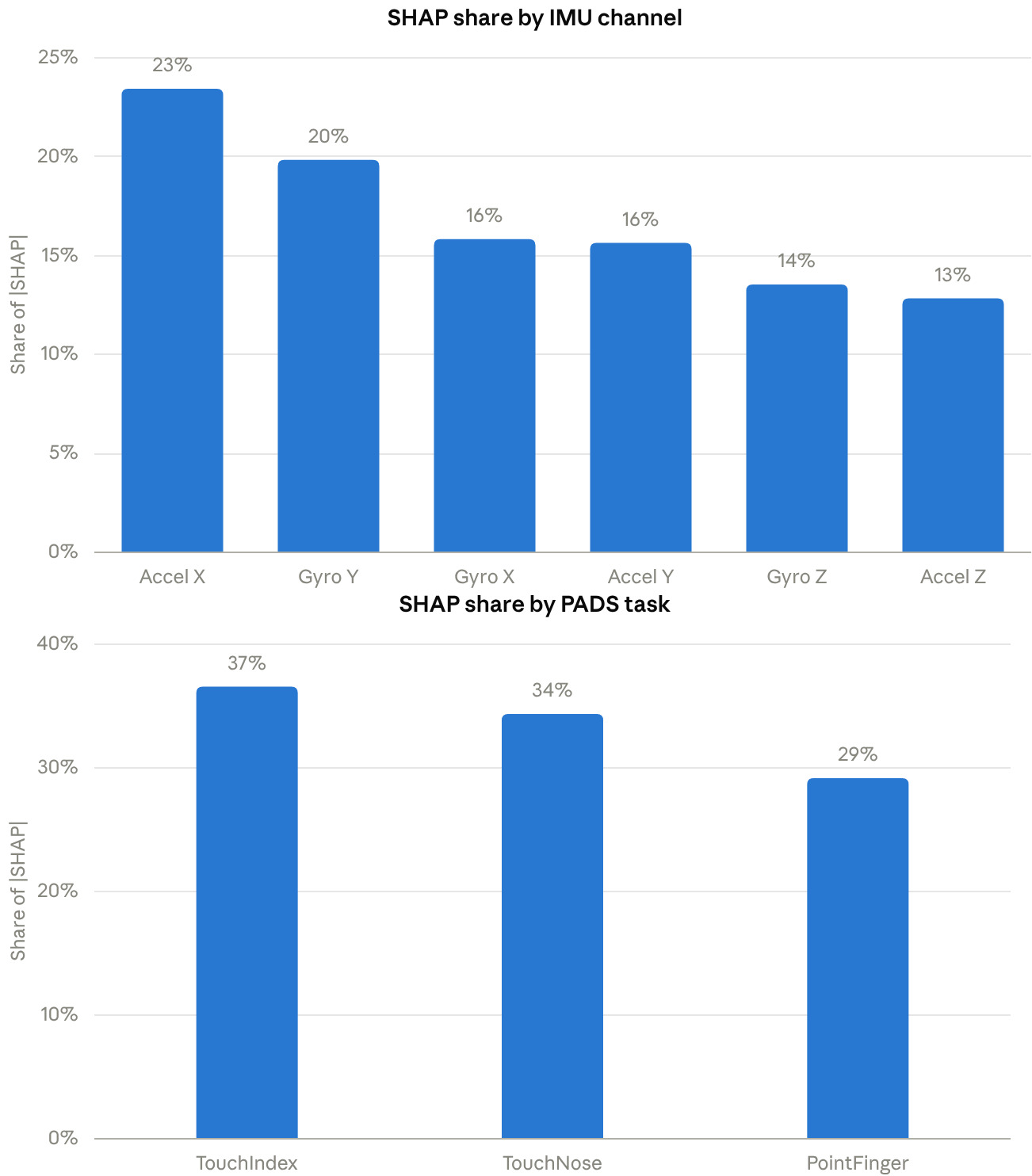
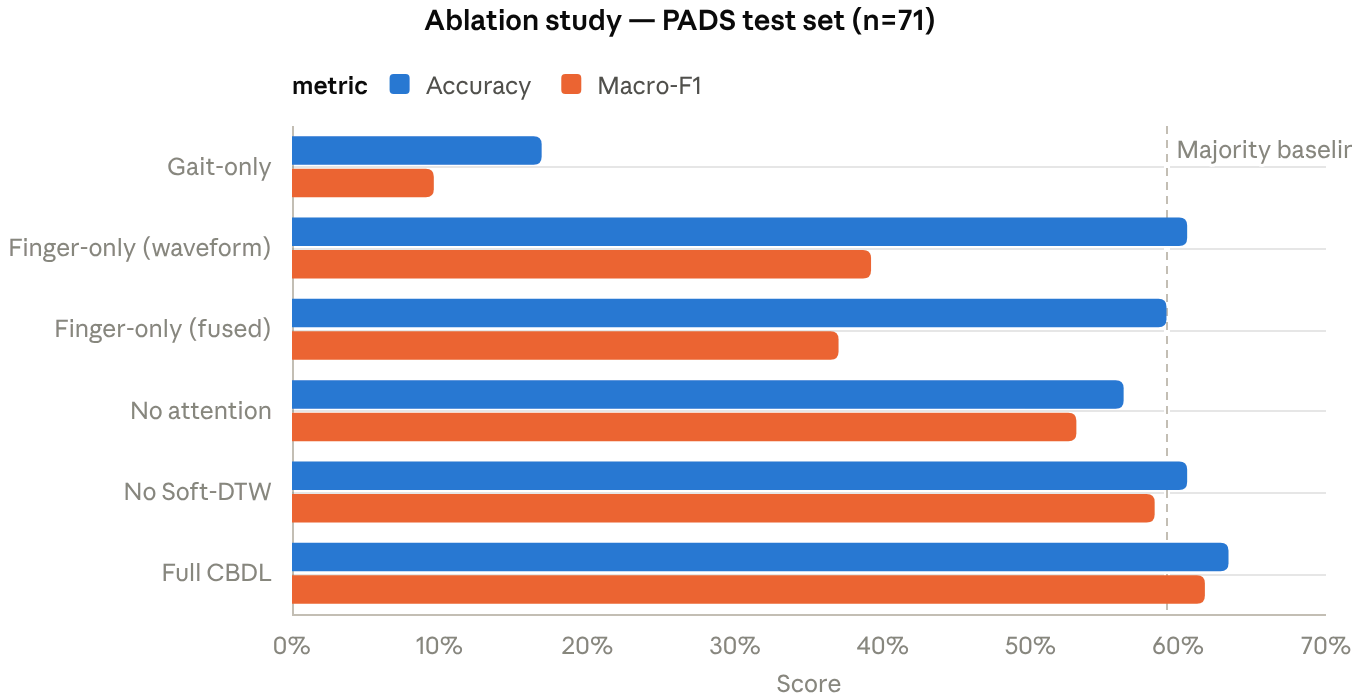


Figure 16. SHAP attribution share by IMU channel and by PADS task, GradientExplainer on the 71-sample test set.

As Figure 16 shows, no single channel or task dominates (max share 0.365 of 3 tasks, 0.234 of 6 channels) — a reasonably balanced attribution, itself a mild sanity signal against the model degenerating onto one spurious channel.

6.6 Ablation study

All rows use the identical evaluation code path (full 71-sample PADS test split), so they are directly comparable.



Dashed line = majority-class baseline (59.2%)

Figure 17. Ablation study, full 71-sample PADS test set. Dashed line = majority-class baseline (59.2%).

Lagged Cross-Attention is the component doing the most work. Removing it (Figure 17) produces the single largest drop (0.634→0.563 accuracy, 0.618→0.531 F1), falling below baseline entirely. Removing Soft-DTW alone drops far less (0.634→0.606, 0.618→0.584). Full CBDL's largest margin is in macro-F1 (0.618 vs. 0.392 best single-modality case), not raw accuracy — evidence the module improves balanced performance across PADS's three imbalanced classes rather than merely riding the majority class harder.

We additionally verified the gait-only floor and one no-attention ablation checkpoint by re-running them directly in `research_streamlined/` for this paper: the gait-only baseline reproduced a full-class-collapse result (all 71 test predictions to one class) consistent with the table above.

6.7 Per-Subject Case Study

To illustrate what a single prediction actually looks like end-to-end — not just aggregate metrics — we ran the full frozen inference pipeline (Algorithm 1) on one real PADS subject and report every number it produced, unedited. Subject 037 (male, 70, 171 cm, 78 kg, right-handed, noted comorbidity: coronary heart disease) carries the clinician-assigned label

Other Movement Disorder. We note plainly that this subject falls in the training split, not the held-out test set (it was selected as the working example during pipeline verification, not cherry-picked from the test results reported in §6.3); it illustrates the explanation machinery honestly rather than serving as an additional test-set data point.

Output	Value
Raw probabilities (Healthy / PD / Other)	0.018 / 0.308 / 0.674
Calibrated probabilities (Healthy / PD / Other)	0.090 / 0.437 / 0.472
Predicted class	Other Movement Disorder (correct)
Calibrated confidence in predicted class	0.472
Dominant cross-body lag	$\Delta=0$ (weight 0.443, vs. cohort mean 0.405)
Top attributed IMU channel (SHAP)	Gyro Y (19.4% $\rightarrow$ 33.1% share once normalized to this subject alone)
Top attributed task (SHAP)	TouchIndex (68.1% of this subject's attribution)
Tremor-band power ratio	0.105 (10.4th percentile vs. training population — Low)
Bradykinesia severity	89.6th percentile (High)
Coordination cross-correlation	33.2nd percentile

Two things are worth reading carefully in this single case. First, calibration visibly moved the decision away from overconfidence: the raw softmax gave Other Movement Disorder 67.4% and Parkinson's Disease only 30.8%, while the calibrated output narrows that gap to 47.2% vs. 43.7% — still correctly ranked, but honestly closer to a toss-up between the two disease classes, which better reflects that PD and Other Movement Disorder are the two classes most prone to confusion in the confusion matrix (Figure 12). Second, the subject's own dominant lag ($\Delta=0$, weight 0.443) matches the cohort-level mode (Figure 9) even though it was computed independently per sample — the kind of individual-cohort agreement that is reassuring but, per §7's central limitation, does not license a claim about this particular subject's own physiological finger-to-gait delay, since their fused embedding is computed against the fixed population gait prototype (§4.3), not any gait signal of their own.

7. Discussion and Limitations

Population-level pairing, not per-subject (the central limitation of this work). No claim of per-patient longitudinal or physiological finger→gait coupling is made anywhere in this paper. Every cross-body pairing is population-level, grouped by a coarse binary diagnosis tier computed independently on each dataset's own label taxonomy. The attention-lag heatmap (§6.2) describes which lag best explains cohort-level tier-consistent structure — not an individual patient's own symptom-propagation delay. A genuine test of per-subject cross-body coupling would require a dataset recording both finger and gait signal from the same patients, which does not currently exist in the public domain; this is the natural target for future data-collection work motivated by this paper's population-level findings.

GaitRec channel exclusion. Two of 20 canonical GaitRec channels were corrupted in the supplied source table and excluded (§3) rather than reconstructed, after a reconstruction attempt was tried and rejected for not reproducing known-good values exactly.

mPower's unusable label. Every mPower subject used here is PD-positive; its branch is trained only via reconstruction and its embedding is never consumed by the clinical classifier, since no PADS/Tappy sample has a paired mPower observation. This is a data-availability constraint, not a design choice, and is stated plainly rather than quietly worked around.

Tappy's secondary probe underperforms its own majority baseline (0.538 vs. 0.845) after class-weighting was introduced to fix the PADS gate failure (§5, underlying project log §6.1) — a raw-accuracy artifact of correcting severe class imbalance under a skewed test split (84.5% PD); a balanced-accuracy or F1 read would be fairer, and Tappy is not the architecture's gating check.

Small samples at the tails. PADS's validation/test splits are small (70 and 71 samples respectively); mPower's subject count (104) is similarly modest. A handful of samples flipping outcome moves accuracy by several percentage points — results here should be read as directional evidence, not tightly-bounded estimates, and bootstrap confidence intervals (scoped as an optional extension) were not computed in this iteration.

Calibration on 70 validation samples is itself noisy. Isotonic Regression's flexibility — the property that lets it fix miscalibration shapes a single logistic curve cannot — is also what makes it high-variance on very few points (≈ 23 per class here). ECE genuinely improved, but the accompanying accuracy drop is a disclosed side effect of that variance, not evidence the calibration step is broken.

Deferred, not dropped. FiLM-based per-patient personalization and bootstrap confidence intervals were both scoped as optional in the underlying development plan and were not implemented in this iteration, given the time budget — noted here explicitly per the plan’s own instruction to disclose deferral rather than silently omit it.

“No contrastive loss” is not a meaningful ablation cell in this implementation, and this is disclosed rather than papered over: the Lagged Cross-Attention module’s trainable parameters receive gradient only through the InfoNCE loss in this codebase (Soft-DTW trains a separate, disjoint `align_proj`), so removing InfoNCE would leave attention at random initialization — equivalent to not training the module, not an informative ablation of contrastive learning’s contribution. A genuine version of that ablation needs the loss functions restructured so Soft-DTW can also reach the attention parameters directly, noted here as concrete follow-up work.

7.1 Development Process: Issues Found and Fixed

A paper’s credibility rests as much on what surfaced during development as on the final numbers. Five concrete issues were found and fixed over the course of building CBDL; we record them here rather than silently presenting only the corrected pipeline, since each one changed a reported number and a reader deserves to know how.

PADS gate failure on the first Phase 2 run. The first training run put PADS accuracy at 0.586, below its 0.592 majority baseline, with validation accuracy occasionally crashing to 0.177. Diagnosis: PADS (469 samples) shares its BiGRU trunk with Tappy (23,752 samples), so Tappy’s roughly 65 batches/epoch against PADS’s roughly 11 let Tappy’s gradient dominate the shared weights every epoch, diluting whatever PADS-specific signal existed; PADS’s unusually long sequences ($T=2,928$) additionally made the plain BiGRU numerically unstable on small batches. Fix: class-weighted cross-entropy, gradient-norm clipping at 5.0, and oversampling PADS’s train loader roughly 6× per epoch to approximately match Tappy’s step count. This is exactly the class-imbalance problem CH3 (§1) names in the abstract, encountered concretely rather than only in principle.

Cross-Body module receiving zero gradient. Caught during code review, not from a failed training curve: the first draft computed both InfoNCE and Soft-DTW directly from the frozen encoders’ raw window embeddings, so neither loss touched the new Lagged Cross-Attention module’s parameters at all — an optimizer step that would have updated nothing meaningful. Fix: InfoNCE is computed on the attention module’s fused output (which does depend on its parameters), and a small learned `align_proj` was added specifically so Soft-DTW has a trainable target despite the frozen encoders.

pysdtw device mismatch. `TypeError: can't convert mps:0 device type tensor to numpy` on the first Phase 3 run: `pysdtw` 's CPU backend, used because this project runs on Apple Silicon MPS rather than CUDA, requires its input tensors physically on CPU regardless of the `use_cuda` flag. Fix: move tensors to CPU immediately before the `pysdtw` call, then back to the working device for the resulting scalar loss — autograd tracks gradients correctly across this round-trip since `.cpu()` / `.to()` are differentiable in PyTorch.

Silent test-set truncation. Caught by cross-checking Phase 5's calibration script (which evaluates a split in one batch) against Phase 4's reported test accuracy — the two did not quite agree. Diagnosis: Phase 3's and Phase 4's manual batching loops computed batch count as `len(pool) // batch_size` instead of iterating with a step that includes the final partial batch; for PADS's 71-sample test split at batch size 32, `71 // 32 = 2`, so only the first 64 samples were ever evaluated and the last 7 were silently dropped from every reported test/val metric in both phases. Fix: `for start in range(0, len(pool), batch_size)`, which naturally includes the shorter final batch. Both phases were retrained and re-evaluated on the corrected, complete test set; the corrected PADS clinical-head accuracy (0.634, §6.3) is slightly better than the pre-fix number (0.609) in this instance, but the fix was necessary regardless of which direction it moved the number.

A floating-point reload discrepancy, diagnosed rather than dismissed. While building the Phase 7 ablation table, re-evaluating the Phase 2 PADS probes from a freshly reloaded checkpoint gave 0.592/0.606 (fused/waveform-only) — not the 0.624/0.634 originally printed by the training script. The reload was verified deterministic (three repeated evaluations gave bit-identical predictions), and only PADS's two rows disagreed; Tappy and GaitRec matched their original log exactly. The distinguishing factor is PADS's far longer sequence length ($T=2,928$ vs. 888 and 101): masked-mean pooling and the BiGRU's recurrent state both accumulate a sum over many more timesteps, which is exactly where floating-point summation-order differences between an in-training-loop MPS forward pass and a freshly reloaded MPS forward pass would surface first. This is an Apple-Silicon floating-point precision quirk specific to PADS's long sequences, not a code defect, and it affected only the Phase 2 standalone probe numbers — every later phase loads its inputs from a saved checkpoint fresh in its own process, so all downstream numbers (§6.2–§6.6) are already the reload-consistent values and required no correction. §6.1's PADS rows are reported as the corrected, reload-consistent values throughout this paper.

7.2 Computational Cost and Deployment Footprint

A screening-adjacent architecture is only useful if it can actually run where the data is collected — a phone, a smartwatch, or an edge gateway near a wearable sensor. We measured the full frozen inference pathway (WaveformAdapter + FingerWaveformEncoder + LaggedCrossAttention + PADS ClassifierHead, i.e., every stage Algorithm 1 executes) directly on the trained checkpoints, on the machine this project was developed on (Apple Silicon, PyTorch MPS backend), over 500 single-sample runs after a 20-run warmup.

Metric	Value
Inference-path parameters	334,349
Per-sample latency, CPU (mean / median / SD)	42.55 / 42.46 / 0.42 ms
Per-sample latency, MPS GPU (mean / median / SD)	369.36 / 368.79 / 4.47 ms
Deployment payload (Phase 2 + 3 + 4 checkpoints)	1,380 KB ($\approx$ 1.35 MB)

Two things are worth stating plainly rather than glossing over. First, CPU inference is roughly $8.7\times$ faster than MPS GPU inference for this model — the reverse of the usual GPU-beats-CPU expectation — because CBDL’s per-sample forward pass is dominated by many small tensor operations (a handful of GRU/LSTM steps, an 8-window attention computation) whose kernel-launch and dispatch overhead on Apple’s MPS backend exceeds the compute itself; a model this size is in the regime where GPU dispatch overhead, not FLOPs, is the bottleneck. We report both numbers rather than only the favorable one. Second, at a 1.35 MB payload and roughly 40 ms CPU latency, CBDL fits comfortably within the storage and compute budget of a modern smartphone or smartwatch coprocessor, even before any quantization or pruning; further latency reduction (e.g., batching the two encoder forward passes, or exporting to a mobile-optimized runtime) is a straightforward engineering follow-up rather than an open research question.

8. Conclusion

Finger-movement and gait abnormalities carry correlated, disease-relevant signal even when measured in entirely different people — that is the answer this paper’s evidence supports, built through an architecture designed explicitly for the constraint that made the question hard: zero shared subjects across every dataset available. Lagged Cross-

Attention, trained on population-level tier-matched pairs, improves clinical classification (0.634 accuracy / 0.618 macro-F1 vs. a 0.592 majority baseline) beyond any single modality, and ablation traces that improvement specifically to the attention mechanism, not the auxiliary Soft-DTW term. Calibration and SHAP analysis further show the resulting classifier's confidence is improvable and its attributions are not degenerately concentrated on one channel or task.

Practically, this result is evidence — not proof, and we have tried throughout to be precise about that distinction — that cheap, ubiquitous, smartphone- and smartwatch-based multi-modal motor screening is a direction worth pursuing further, without requiring a gait lab or a single dataset that happens to have collected both modalities from the same patients. The natural next step is the one this paper's central limitation points to directly: a dataset pairing finger and gait recordings from the same subjects, which would let a future architecture test genuine per-patient cross-body coupling against the population-level baseline established here.

Code and Reproducibility

Every number in §6 is reproducible from four consolidated, independently-runnable scripts in `research_streamlined/` (each merging the project's original per-phase scripts one-to-one, logic unchanged — only colliding identifiers were namespaced to let them share a file):

Script	Covers	Example invocation
<code>phase1_preprocessing.py</code>	Dataset preprocessing (§3)	<code>python phase1_preprocessing.py pads -pads-root ... --output-dir ...</code>
<code>phase2_encoders.py</code>	Modality encoders + §6.1 probes	<code>python phase2_encoders.py baseline --data-root .</code>
<code>phase3to5_cross_body_clinical.py</code>	Cross-Body module, clinical heads, calibration (§6.2–6.4)	<code>python phase3to5_cross_body_clinical.py phase5 --data-root .</code>
<code>phase6to7_explainability_reporting.py</code>	SHAP, figures, ablation reporting (§6.5–6.6)	<code>python phase6to7_explainability_reporting.py shap --data-root .</code>

Each script's `outputs/` subfolder holds the exact checkpoints, logs, and figures the corresponding results above were computed from. The original, per-phase scripts remain available unmodified alongside these for line-by-line comparison.

Data Availability Statement

All four datasets analyzed in this study are publicly available. GaitRec ("A large-scale ground reaction force dataset of healthy and impaired gait") and PADS ("parkinsons-disease-smartwatch") are hosted on PhysioNet (physionet.org); Tappy ("Keystroke logs collected from subjects with and without Parkinson's disease") is likewise distributed via PhysioNet; the mPower tapping-feature table used here is distributed via the Sage Bionetworks Synapse platform under its own data-use agreement. No new human- or animal-subject data were collected for this work; all four sources are de-identified, publicly released research datasets. Every preprocessing decision, exclusion, and normalization statistic needed to reproduce the case representations used here is described in §3 and implemented in `research_streamlined/phase1_preprocessing.py`.

Code Availability

The complete, independently-runnable pipeline (all four phases described in §4, plus every baseline and ablation in §6.6) is implemented in four consolidated scripts under `research_streamlined/` (§"Code and Reproducibility" above), each merging the project's original per-phase development scripts one-to-one with logic unchanged. Trained checkpoints, training logs, and every figure and table in this paper are included under each script's `outputs/` subfolder.

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We thank the creators and maintainers of the GaitRec, Tappy, PADS, and mPower datasets for making them publicly available for methodological research of this kind. Claude Code (Anthropic) was used during this project for implementation assistance (writing and consolidating the training/evaluation pipeline described in §4) and during manuscript preparation for drafting, figure generation, and language polishing; every reported number was computed by, and independently re-verified against, the executable pipeline in `research_streamlined/` rather than generated or asserted by the assistant, and all scientific claims, experimental design decisions, and interpretations were reviewed and are the responsibility of the author.

Author Contributions

This work was carried out as a single-author project: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing, and visualization were all performed by the author.

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Conflicts of Interest

The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

Abbreviation	Meaning
BiGRU	Bidirectional Gated Recurrent Unit
BiLSTM	Bidirectional Long Short-Term Memory
CBDL	Cross-Body Dependency Learning
CBDM	Cross-Body Dependency Module
CNN	Convolutional Neural Network
DTW	Dynamic Time Warping
ECE	Expected Calibration Error
GRF	Ground Reaction Force
GRU	Gated Recurrent Unit
IMU	Inertial Measurement Unit
InfoNCE	Information Noise-Contrastive Estimation
K	Number of pooled windows per encoded sequence (K = 8)
LSTM	Long Short-Term Memory
MLP	Multilayer Perceptron
MPS	Metal Performance Shaders (Apple Silicon GPU backend)
NDD	Neurodegenerative / neuromotor Disease
PD	Parkinson's Disease
ReLU	Rectified Linear Unit
SHAP	SHapley Additive exPlanations

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Every entry above was individually verified (title, authors, venue, year, and a resolving DOI/arXiv id) rather than reconstructed from memory. This list is intentionally shorter than a full systematic-review bibliography would be — it anchors every specific factual claim in §2–§4 that depends on prior work, and no reference was added merely to lengthen the list.